

cloud → store data, run software and use computing resources (like server, storage and applications)

characteristics → 1) on-demand 2) broad network access 3) Resource pooling
4) scalability & elasticity 5) pay-as-you-go model 6) high availability

Application → 1) data storage and backup 2) web hosting 3) online education 4) business app
5) software development and testing 6) entertainment services

Advantages → 1) cost effective 2) easy access 3) scalable 4) automatic updates
5) data backup & recovery.

Disadvantages → 1) internet dependency 2) security issues 3) limited control 4) downtime
5) data privacy concerns.

feature	cloud	Grid
Purpose	Provide services like storage, software and servers on demand	solves large computational problems using many computers.
Resource ownership	owned and managed by a single cloud provider	belongs multiple organizations
Access method	Access through using internet browser or API	accessed through special grid middleware
Scalability	Highly scalable and elastic	limited scalable, not fully elastic
Cost model	pay-as-you-use pricing	no direct cost model
Ease of use	very easy user friendly	complex to configure and use
Resource Allocation	dynamic and automatic	static or preplanned.
Availability	high availability with backup and redundancy	availability depends on participating nodes.
security management	centralized security by provider	Distributed and complex security
Examples	AWS, Microsoft Azure, Google cloud	SETI@home, scientific research grid

feature	centralized	Decentralized
control	All control one central authority	control distributed multiple nodes
Decision making	Decision are taken one central point	Decision are taken by multiple entities
Failure handling	single point of failure - if central server fails, whole system stops.	No single point of failure, system continues to work
Scalability	limited	highly scalable
Performance	can become slow with heavy load	Better performance due to load sharing
Ex	Traditional DB, mainframe systems	Blockchain, Peer-to-peer networks

virtualization → one physical hardware run multiple VM at the same time by creating virtual version of hardware, operating system, or resources.

Types of virtualization → 1) hardware/server 2) OS 3) storage 4) Network 5) Desktop

Characteristics → 1) Abstraction 2) Isolation 3) Encapsulation 4) Resource sharing
5) hardware independence 6) scalability & elasticity 7) security
8) disaster recovery

Application → 1) cloud computing 2) server colidation 3) software testing and development
4) Education and training 5) Running multiple OS

Advantages → 1) Better Resource utilization 2) cost saving 3) scalability 4) Isolation
5) Easy maintenance.

Working → 1) A software called hypervisor is installed on physical hardware
2) the hypervisor divides physical resources (CPU, RAM, storage)

3) these resources are assigned to virtual machines (VMs)
4) Each VM runs its own operating system independently

5) All VMs work together on the same physical machine.

Hypervisors → Also called VMM that creates, manages and runs virtual machines (VMs) on physical hardware

functions → 1) creates VMs 2) Resource management 3) Isolation 4) migration
5) Snapshots and cloning

Types → Type 1 (Bare metal) → Run directly on physical hardware no underlying OS needed
eg → VMware ESX, Microsoft Hyper-V, Xen

Type 2 (Hosted) → Runs on top on an existing OS, eg → VMware Workstation, Virtual Box

Importance of hypervisor's → 1) foundation of virtualization 2) scalability
3) Isolation and security 4) Efficient Resource utilization 5) disaster recovery

Examples → 1) AWS EC2 → uses hypervisors to run virtual instances for customers
2) Microsoft Azure VMs → Hypervisors manages VM provisioning and scaling

3) VMware vSphere/ESXi → Enterprise-grade bare-metal hypervisor for data centers

4) Oracle VirtualBox/VMware Workstation → Type 2 hypervisors for personal use and test

	single tenancy	multi-tenancy
feature definition	one application instance is used by only one user/organization	one application instance is shared by multiple users/organizations
Resource usage	dedicated resources for each tenant	Resources are shared among tenants
Cost	more expensive due to dedicated infrastructure	cost-effective due to shared resources
security	Higher security and data isolation	security depends on logical isolation
customization	High level of customization possible	Limited customization.
Scalability	less scalable and harder to manage	Higher scalable and easy to manage
Example	private cloud, dedicated hosting	SaaS platforms like Gmail, Salesforce.

multi-tenancy $\Rightarrow$ multi-tenancy is ~~an~~ single application instance is shared by multiple users (tenants)

Types $\Rightarrow$ 1) database-level 2) schema-level multi-tenancy 3) Application-level
 1) virtual machine-level
 Application $\Rightarrow$ 1) cloud computing 2) software as a service (SaaS) 3) web hosting platform
 1) Email services 5) CRM and ERP systems 6) online learning platforms

Advantages $\Rightarrow$ 1) cost effective 2) scalability 3) efficient resource utilization
 1) Easy maintenance 4) better performance

Disadvantages $\Rightarrow$ 1) security Risk 2) limited customization 3) performance issues during heavy load
 1) complex Architecture 5) dependency on provider

Working $\Rightarrow$ 1) multiple tenants access same application
 2) system identifies tenant using login or tenant ID
 3) Data is stored and retrieved separately for each tenant
 4) Resources (CPU, memory, storage) are shared efficiently
 5) Each tenant feels like they are using their own system

API $\Rightarrow$ An API is a set of rules and protocols that allows different software application to communicate with each other.

Role of API $\Rightarrow$ 1) services Access 2) Automation 3) integration 4) integration
 5) platform independence

Types $\Rightarrow$ 1) REST APIs 2) SOAP APIs 3) RPC APIs 4) SDK APIs

API work $\Rightarrow$ 1) Request (Application $\rightarrow$ service) 2) cloud process the request and perform operation
 3) Response is sent back to the client application.

Benefits $\Rightarrow$ 1) Automation 2) integration 3) scalability 4) flexibility 5) Efficiency

Billing and metering $\Rightarrow$ tracking, measuring and charging of cloud resources used by customers

Features $\Rightarrow$ 1) Pay-as-you-go 2) Resource monitoring 3) Transparency 4) cost optimization
 eg $\Rightarrow$ 1) AWS EC2 2) Azure storage 3) Google cloud functions

Cloud management $\Rightarrow$ monitoring, controlling and optimizing cloud resources ensures performance, security, and cost-effectiveness

Features $\Rightarrow$ 1) Resource provisioning 2) monitoring & performance 3) security & compliance
 4) cost management 5) Backup & Disaster Recovery.

eg $\Rightarrow$ 1) AWS Cloudwatch 2) Azure monitor 3) Google cloud operations

Cloud Tooling $\Rightarrow$ software tools and platform provided by cloud providers to manage, monitor, and optimize cloud resources

Features $\Rightarrow$ 1) Resource management tools 2) monitoring & analytics tools
 3) security tools 4) cost & Billing tools.

eg $\Rightarrow$ 1) AWS management console 2) Azure Portal 3) Google cloud console 4) Terraform

Cloud Automation $\Rightarrow$ use of software or scripts to perform tasks automatically without human intervention.

Features $\Rightarrow$ 1) Provisioning Automation 2) Auto-Scaling 3) Backup & Recovery
 4) monitoring & Alerts 5) Deployment Automation

eg $\Rightarrow$ 1) Infrastructure provisioning 2) continuous Deployment
 3) Auto Scaling 4) Backup Automation

Types of VMs $\Rightarrow$

1) System VM $\Rightarrow$ Provide complete virtual platform that supports the execution of a full operating system
 features $\Rightarrow$ 1) virtualizes the entire hardware 2) Allows multiple OS in single physical machine
 3) managed by hypervisors (Type 1, or 2) eg $\Rightarrow$ VMware ESXi, Hyper-V, KVM, Virtual Box, X

use case $\Rightarrow$ 1) server consolidation 2) cloud hosting 3) Running multiple OS simultaneously
 2) Process VMs $\Rightarrow$ Provide run single application independent of the underlying machine or OS
 features $\Rightarrow$ 1) short lifespan 2) provide platform independent for application $\Rightarrow$ eg $\Rightarrow$ JVM, Py, vir
 use $\Rightarrow$ code execution environment 3) sand box application 4) V8 engine for JS

IaaS → virtualized, provide virtualized computing resources, users can rent servers, storage, networking and other infrastructure

Features → 1) virtual machine 2) storage 3) networking 4) pay-as-you-go 5) scalability 6) Automated provisioning.

Working → 1) requests resources 2) allocates virtual resources, 3) install manages OS, software, applications 3) billing based on (CPU hours, storage used, network traffic)

Advantages → 1) cost-efficient 2) flexible & scalable 3) quick deployment 4) control 5) disaster recovery.

Disadvantages → 1) management responsibility 2) security risk 3) complexity

Eg → 1) AWS 2) Azure 3) VCP

Use Case → 1) startups 2) website hosting 3) disaster recovery 4) Big data processing.

PaaS → provides Ready-to-use platform for developing, testing and deploying applications user do not manage underlying infrastructure (servers, storage, or networking)

Features → 1) Development tools 2) middleware 3) DB management 4) scalability 5) integrated development environment (IDE) 6) multi-user collaboration.

Working → 1) sign up 2) allocates development platform (OS, middleware, database, runtime) 3) user build and deploy applications 4) ~~manages~~ ^{provider} manages underlying hardware, networking, OS 5) Billing is Based on usage (Power, storage, API calls)

Advantages → 1) faster development 2) cost-efficient 3) scalable 4) focus on development 5) collaboration 6) maintenance-free

Limitations → 1) less control 2) vendor lock-in 3) limited flexibility. 4) AWS → Elastic

Eg → 1) VCP 2) Microsoft Azure 3) Heroku (supports multiple languages)

Use Case → 1) Development 2) API development and integration 3) collaborative 4) Analytics and Business Intelligence Apps.

SaaS → Provide Ready-to-use software application, easy access ~~without~~ ^{without} worrying installation, maintenance or infrastructure.

Feature → 1) Accessibility 2) Automatic updates 3) multi-tenant architecture 4) subscription-based 5) scalability 6) integration.

Working → 1) sign up 2) provider hosts the software on cloud server 3) user access via Net. 4) provider manages everything 5) Billing is usually subscription-based

Advantages → 1) No installation required 2) Automatic updates 3) cost-efficient 4) accessible anywhere 5) scalable 6) maintenance-free.

Disadvantages → 1) Internet dependency 2) limited control 3) data security concerns 4) vendor-lock in

Eg → 1) Email & collaboration 2) Productivity software 3) CRM software 4) Accounting 5) storage.

Use Case → 1) Remote work 2) Business Application 3) Education 4) Data storage.

Types of deployment model

1) Public → where service and infrastructure own, managed by third party APP, delivered to the public

Eg → AWS, VCP, Azure

Characteristics → 1) shared infrastructure 2) pay-as-you-go pricing 3) Accessible anyone

Advantages → 1) low cost 2) No setup & maintenance 3) high scalability 4) quick deployment

Disadvantages → 1) lower security 2) less customization 3) compliance issues.

2) Private → dedicated single organization, infrastructure managed internally by third party architecture

Characteristics → 1) single-tenant environment 2) high control and security 3) customizable & efficient

Advantages → 1) high security & privacy 2) full control 3) customization 4) support for legacy systems

Disadvantages → 1) cost 2) limited scalability 3) complex setup & maintenance.

3) Hybrid → Public + private, sensitive operation → private cloud and public cloud for scalability and cost

Characteristics → 1) Best of both worlds 2) cost optimization 3) improved security 4) flexibility.

Advantages → 1) mix of on-premise and cloud services 2) data portability between environments 3) flexible resource planning.

Disadvantages → 1) complex management 2) latency issues 3) high initial setup cost.

4) Community cloud → shared by multiple organizations that have similar goals, regulations or security requirements.

Characteristics → 1) collaborative environment 2) shared cost and infrastructure 3) common policies and compliance

Advantages → 1) cost sharing 2) Better security 3) Easy collaboration 4) customized for community needs

Disadvantages → 1) limited scalability 2) customization restriction 3) complex governance.

AWS → provides IaaS, PaaS, SaaS with pay-as-you-go pricing model

Service → 1) compute services → EC2 → virtual server, Lambda → serverless computing, Elastic Beanstalk → development
 2) storage services → S3 → object storage, EBS → Block storage, Glacier → Archival storage
 3) database services → RDS → Relational DB service, DynamoDB → NoSQL DB, Aurora → high performance DB
 4) Networking services → VPC → virtual private cloud, Route 53 → DNS services, ELB → Elastic cloud balancing
 5) Security services → IAM, AWS Shield → DDoS protection, AWS WAF → web app firewall

Features → 1) Scalability 2) High Availability & Reliability 3) security 4) pay-as-you-go pricing
 5) Disaster Recovery 6) cost optimization tool 7) global infrastructure

Advantages → 1) High scalability 2) Flexibility 3) wide range of services
 4) strong security & compliance 5) reliable global infrastructure
 6) cost effective for all business sizes

Disadvantages → 1) complex pricing model 2) difficult for beginners 3) Risk of high cost if not managed properly

Application → 1) hosting 2) data storage & backup 3) Big data analytics 4) IoT & AI Applications

cloud security → refers to technology, policies and controls used to protect data, applications and services in a cloud environment.

Key points → 1) Protect data, applications and infrastructure
 2) Uses encryption to secure data at rest and in transit
 3) Implement Identity IAM
 4) Uses firewalls, VPNs, and intrusion detection system
 5) Ensures data privacy and compliance with law and standards
 6) Follows a shared responsibility model between user and cloud provider

cloud bridge → connects different security layers and environments to ensure continuous protection.

Key points → 1) act as a link between on-premise and cloud security
 2) integrates network, application and data security
 3) helps in secure data transfer between systems
 4) reduce security gaps across platform
 5) ensures consistent environment of security policies

cloud management → involves planning, implementing, monitoring and controlling and analyzing cloud security policies

Key points → 1) defines security policies and rules 2) performs risk management and threat analysis
 2) manages user roles and access rights 3) monitors systems using logs and audits
 3) handles incident detection and response 4) ensures regular updates and security patches

cloud reference model → provides a structured framework to understand and implement cloud security.

Key points → 1) Confidentiality 2) Integrity 3) Availability 4) Authentication
 5) Authorization 6) Accountability